RACE: Scalable Statistical Estimation of Functional Consistency in LLM Neurons

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Stable residual alignment reveals neurons for **specific model behaviors.**

01 Collect Evidence from Residual Stream

Existing attribution approaches face a severe trade-off between computational tractability and functional validity. On one hand, gradient-based methods require iterative backward passes that incur prohibitive overhead, limiting scalable domain-wide auditing. On the other hand, raw activations fail to measure functional consistency due to three key flaws: (1) Lack of direction; (2) Orthogonal cancellation; and (3) Masked instability.

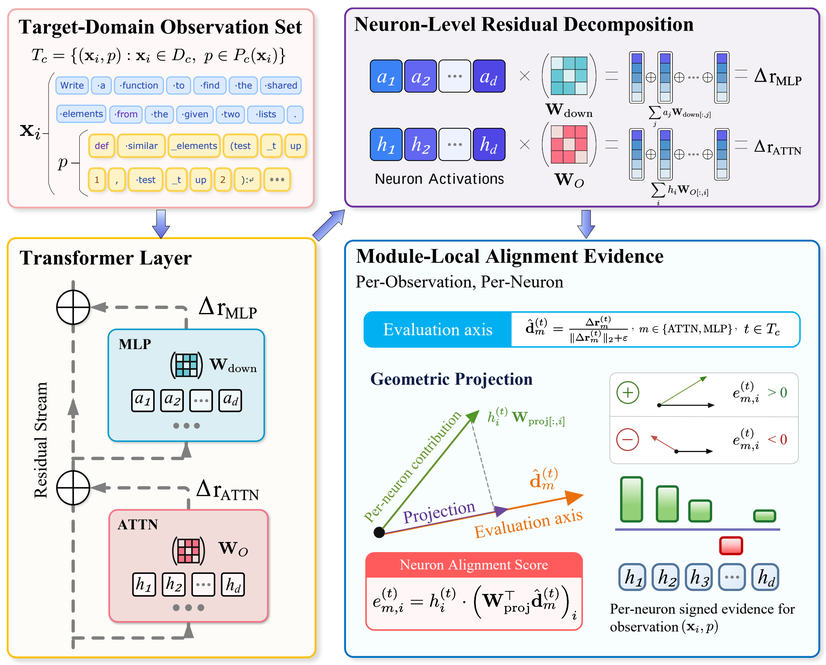
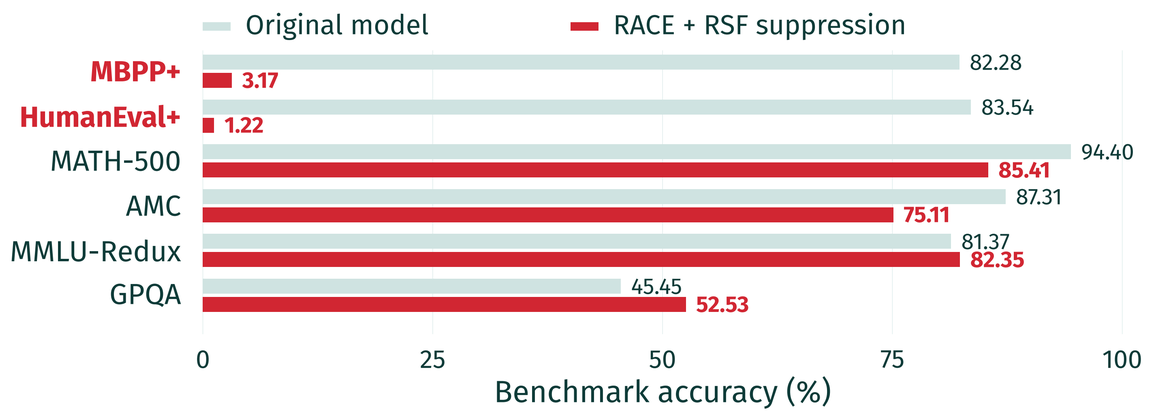


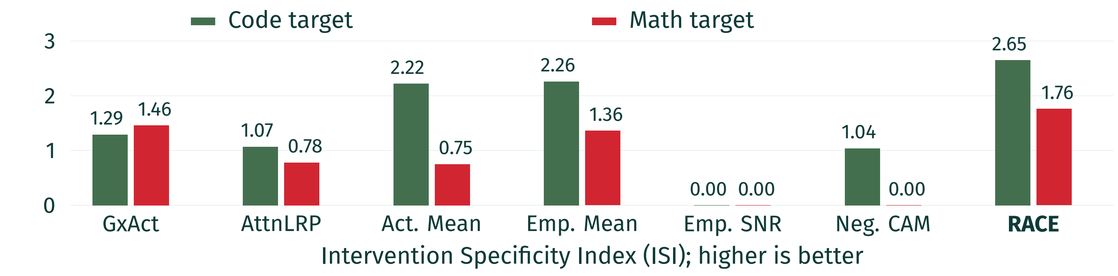
Figure 1: Residual-Direction Alignment (RDA) at forward-pass cost. Each module update is decomposed into rank-1 neuron writes and projected onto the realized update direction $\hat{d}$. The resulting scalar e_j provides signed, module-local evidence: positive values reinforce the module's update, while cancelling orthogonal components are discarded.

03 Selective Suppression: Broad to Narrow

Code-selected MLP neurons: strong drops on the target benchmark and the same-domain out-of-distribution (OOD) benchmark (HumanEval+ for code, AMC for math), with much smaller effects on non-target benchmarks.



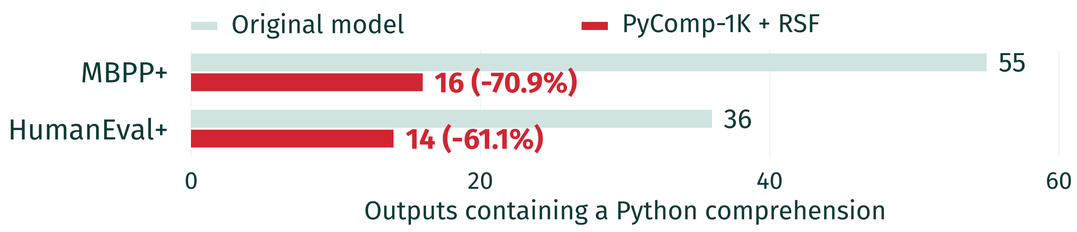
Qwen3-4B-it; MBPP+ scoring, WikiText-2 RSF, top-1% MLP suppression per layer. Accuracy is averaged over three runs; code targets are red.



Same top-1% MLP budget with RSF. Intervention Specificity Index (ISI) $= \max\{0, \log(\Delta_T/(\Delta_G + \delta))\}$, where Δ_T and Δ_G are the average relative accuracy drops on target-domain (incl. OOD) and general benchmarks. Higher ISI indicates stronger target-domain suppression with minimal non-target degradation.

From Broad Capability to a Narrow Behavior

PyComp-1K: 1,000 AST-verified Python statements targeting comprehensions. Suppress $K_{\text{sel}} = 10$ neurons per layer after RSF.



Qwen3-4B-it; number of outputs containing at least one comprehension.

Functional correctness on the **remaining comprehension-using solutions**: MBPP+ 92.73% → 87.50%; HumanEval+ 86.11% → 85.71% (conditional subset scores).

02 Estimate, Rank, and Filter

Goal: Evaluating Functional Consistency

A domain-specific neuron must satisfy two desiderata across the observation set T_c:

- **Magnitude:** $\mathbb{E}_{t \in T_c}[e_j^{(t)}]$ is large and positive relative to peers in the module.
- **Stability:** $\mathrm{Var}_{t \in T_c}[e_j^{(t)}]$ is small—not driven by rare outliers or sign fluctuations.

Estimate | Bayesian Evidence Aggregation

Motivation: Simple empirical averages discard cross-input variation and obscure sample-to-sample stability patterns.
NIG Working Model: Treat signed RDA evidence as noisy draws $e_j^{(t)} \sim \mathcal{N}(\mu_j, \sigma_j^2)$ with a conjugate Normal-Inverse-Gamma prior $\mathrm{NIG}(\mu_0, \lambda_0, \alpha_0, \beta_0)$.
O(1) Streaming Inference: Maintaining empirical mean $\bar{e}_j$ and centered sum of squares SS_j yields a closed-form marginal Student's t-posterior for the mean evidence:

$$\mu_j \mid \{e_j^{(t)}\} \sim t_{2\alpha_n}(\mu_{n,j},\ \sigma_{\mu,j}), \qquad \text{where} \quad \sigma^2_{\mu,j} = \frac{\beta_{n,j}}{\alpha_n \lambda_n}.$$

The scale $\sigma_{\mu,j}$ captures epistemic uncertainty—inflated by scarce, noisy, or sign-fluctuating evidence.

Rank | Conservative Alignment Magnitude (CAM)

Motivation: Weak or sparse activations can artificially inflate empirical SNR, while global sorting unfairly favors late layers where updates align closer to logits.
Credible Lower Bound: CAM scores neurons by a one-sided conservative $(1-\gamma)$-lower credible bound on positive mean evidence:

$$\mathrm{CAM}_j(\gamma) = \mathbf{1}[\mu_{n,j} > 0] \cdot \max(0,\ \mu_{n,j} - t_{1-\gamma,\,2\alpha_n} \cdot \sigma_{\mu,j})\,.$$

Stratified Budget: Neurons are ranked by CAM within each layer and module to preserve coverage across the model's hierarchical computation.

Filter | Reference-Set Filtering (RSF)

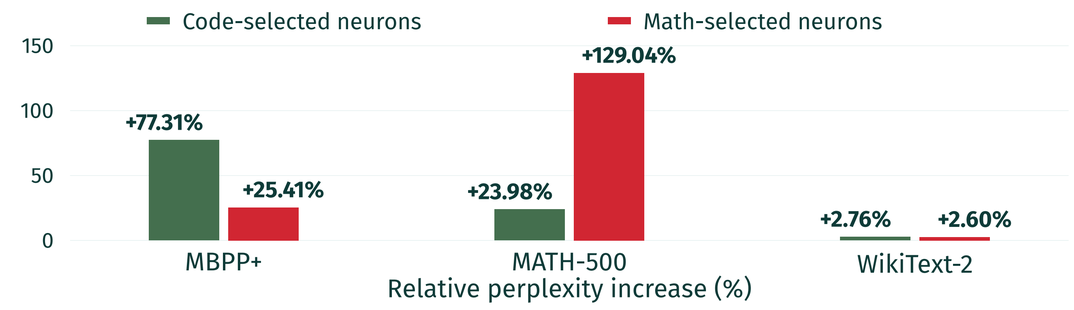
Motivation: Under feature superposition and polysemanticity, raw target rankings conflate domain-specific functions with broadly shared general language capabilities.
Dual-Corpus Filtering: Traverses target ranking Π^ℓ_{tar} and discards neurons that also rank highly on a general reference corpus (WikiText-2, B^ℓ_{ref}) to retain K_{sel} domain neurons:

$$\Pi^\ell_{\text{RSF}} = \left[j \in \Pi^\ell_{\text{tar}} : j \notin B^\ell_{\text{ref}}\right], \qquad S^\ell_{\text{spec}} = \mathrm{First}_{K_{\text{sel}}}(\Pi^\ell_{\text{RSF}}).$$

Prevents baseline linguistic collapse during subsequent behavioral interventions.

04 Verify the Distributional Effect

Target-corpus perplexity rises sharply; the general reference corpus changes by less than 3%.



Qwen3-4B-it, RSF, top-1% MLP suppression per layer. Relative perplexity change; each color uses its own target scoring set.

The Effect Extends to 32B Models

OLMo-3.1-32B-it, math scoring + WikiText-2 RSF; top-5% MLP suppression per layer.

Benchmark	Original	Suppressed
MATH-500 (target)	86.40	**6.72**
AMC (OOD)	79.85	**7.46**
GPQA	48.60	37.79
MMLU-Redux	84.70	77.79

Paper, Code, and Contact

Paper Code

Paper: arXiv:2608.24758v2 **Code:** github.com/Nexround/RACE
Contact: 2430310032@stmail.ntu.edu.cn *pingpeng@ntu.edu.cn